# SCSE 🏛️

## Stablecoin Clearing & Settlement Engine

**Sivasubramanian Ramanathan**
*Product Owner | Fintech & Innovation*
*Ex-BIS Innovation Hub Singapore*

🌏 **Seeking Opportunities in Singapore**
I am looking for roles in **Product Management, Fintech, Payments, RegTech**, and **Digital Assets**.

"I am not just a Product person. **I build.**"

I have worked across product delivery, user research, and cross-agency collaboration. I enjoy solving complex problems and bringing structure to early ideas.

**I care deeply about building products that create real impact.**

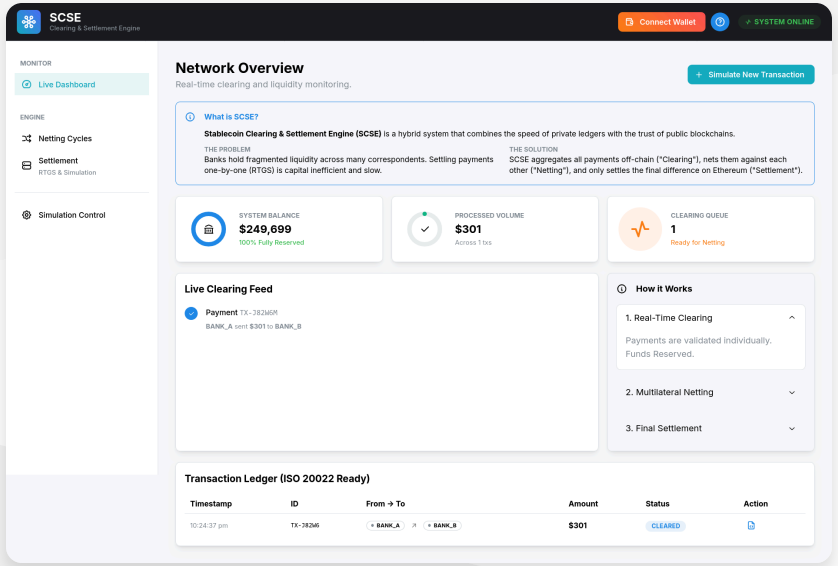
# The Learning Journey: SCSE 🚀

**Goal**: To build a real-world Defi application from scratch to learn full-stack Web3 development.

## Live Dashboard

I built this React workstation to visualize the invisible logic of clearing engines.

- **Concepts Learned**: State Management (Zustand), React Hooks, Wagmi integration.
- **Outcome**: A working simulation of "Real-Time Gross Settlement" vs "Netting".



# My First Smart Contracts 📜

I moved beyond theory and deployed my first Solidity contracts to **Sepolia Testnet**.

## 1. Settlement.sol

- **Function**: `settleBatch(debtors, amounts, creditors, amounts)`
- **Logic**: Atomically collects from debtors and pays creditors. Reverts if *any* transfer fails (All-or-Nothing).
- **Security**: `onlyOwner` modifier ensures only the Clearing Engine can trigger movement.

## 2. MockUSDC.sol

- **Standard**: ERC-20 implementation using **OpenZeppelin**.
- **Feature**: Added a public `mint()` function to simulate a faucet for testing.

> ***Tools Used***: *Hardhat (Compiling/Deploying), Alchemy (RPC), Etherscan (Verification).*

# The Problem: Digital Money is Siloed

In my work analyzing **Cross-Border Payments & CBDCs**, I've seen how liquidity fragmentation kills efficiency.

1. 🔴 **Trapped Liquidity**:
   Stablecoins sit idle in different chains and wallets, requiring massive pre-funding.
2. 🔴 **Slow Finality**:
   Traditional blockchains (15s - 10 min) are too slow for high-frequency point-of-sale payments.
3. 🔴 **High Gas Costs**:
   Settling *every* coffee purchase on-chain is inefficient.

> “***Goal**: Build a **Hybrid Infrastructure** that combines database speed with blockchain finality.*”

# Definitions: Clearing vs Settlement 📖

## 1. Clearing (Off-Chain) ⚡

- **What**: Calculating obligations.
- **Action**: "I owe you $50".
- **Process**:
  - Validation & Compliance (Travel Rule).
  - Internal Ledger Updates.

## 2. Settlement (On-Chain) ⛓️

- **What**: Extinguishing obligations.
- **Action**: "Here is the $50 token".
- **Mechanism**: **ERC-20 Transfer**.
  - Blockchain finality = Settlement.

> *"**Why separate them?**: Netting off-chain turns 1,000 payments into 1 Settlement Transfer."*

# The Problem: The Liquidity Trap 🤔

*Question: "Stablecoin settlement is just a transfer, why build a Clearing Engine?"*

**Answer:** If you settle every payment individually (Gross Settlement), you need massive pre-funded capital.

## Scenario: Gross Settlement (No Engine)

- Alice sends **$1M** to Bob (Needs $1M pre-funded).
- Bob sends **$1M** to Alice (Needs $1M pre-funded).
- **Total Liquidity Locked: $2M**.

> *"Inefficiencies like this are why traditional finance uses Clearing Houses."*

# The Solution: Netting & Stablecoins 💡

The Engine calculates the net obligations *before* moving any tokens.

## Scenario: With Clearing Engine

- Engine calculates: Alice +$1M, Bob -$1M. THEN Bob +$1M, Alice -$1M.
- Net Obligation: **$0**.
- **Total Liquidity Locked: $0**.

> ***"Where is the Stablecoin part?"***
> *In a perfect circle ($0 net), the Stablecoin is the **Unit of Account** (Measuring debt).*
> *In a partial circle (e.g. $1M vs $900k), the **Residual Difference ($100k)** is settled on-chain.*

# The Deep Dive: Identity vs Value 🔍

The core architecture challenge in compliant crypto-finance is separating **Who** from **What**.

## Identity (Off-Chain)

- **Standards**: IVMS101 (Travel Rule).
- **Carrier**: Encrypted P2P Messaging (VASP-to-VASP).
- **Content**: PII, Sanctions Data, Source of Funds.
- **Why?**: Privacy (GDPR) & Scalability.

## Value (On-Chain)

- **Standards**: ERC-20 / Native Token.
- **Carrier**: Public/Private Blockchain.
- **Content**: Amount, Addresses, Proof of Ownership.
- **Why?**: Finality, Trustlessness, Immutability.

> *"**Concept**: The Blockchain settles value; VASPs exchange identity off-chain."*

# Why not just ISO 20022? 🏛️

ISO 20022 is the gold standard for banking (SWIFT MX), but it has limitations in the crypto ecosystem:

1. **Bank-Centric Model**: Assumes counterparties are regulated FIs (BICs) with accounts. In crypto, "Self-Custody" (Unhosted Wallets) exists.
2. **Privacy Leaks**: Rich data payloads (Remittance Info) on a transparent blockchain = Doxing users.
3. **Complexity**: ISO messages are heavy. Crypto needs lightweight, purpose-built identity packets.

## The Hybrid Compromise

- **Internal Ops / Fiat Legs**: Use **ISO 20022**.
- **VASP-to-VASP Messaging**: Use **IVMS101**.

# IVMS 101: The Identity Standard

**IVMS 101** (InterVASP Messaging Standard) is the data model for the "Travel Rule".

- It is **NOT** a transport protocol (it's the payload, not the pipe).
- It is **NOT** XML/JSON specific (though usually JSON).

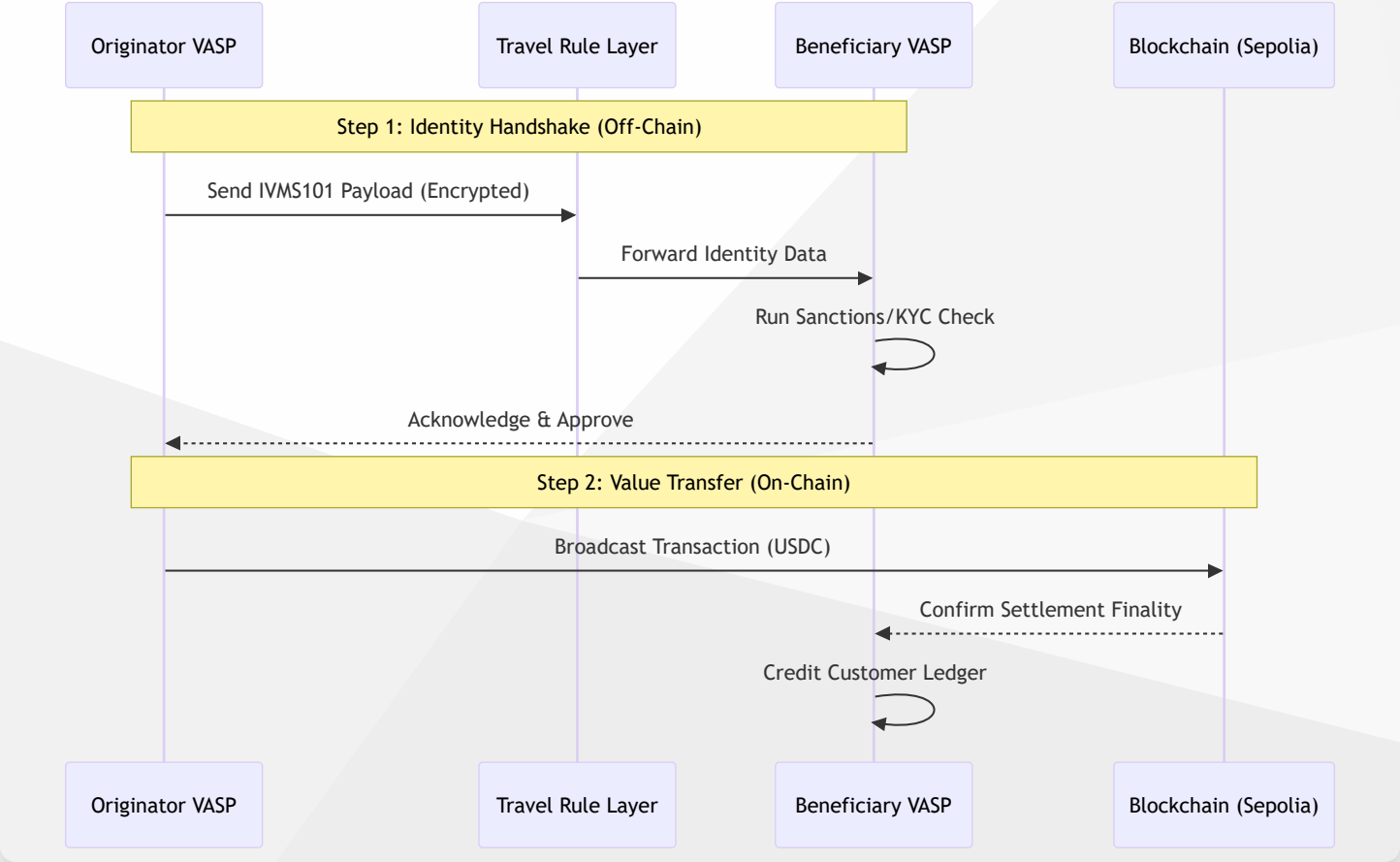
## What's Inside?

- **Originator**: Name, Account Number (Wallet Address), Physical Address, National ID.
- **Beneficiary**: Name, Account Number.
- **Beneficiary VASP**: Identification code (LEI, BIC).

> *"**Analogy**: IVMS101 is the "Envelope" containing the ID cards. The Blockchain is just the "Truck" moving the cash."*

# The Compliant Flow (Custodial)

How a compliant stablecoin transfer *actually* happens.



# The Solution: SCSE 🚀

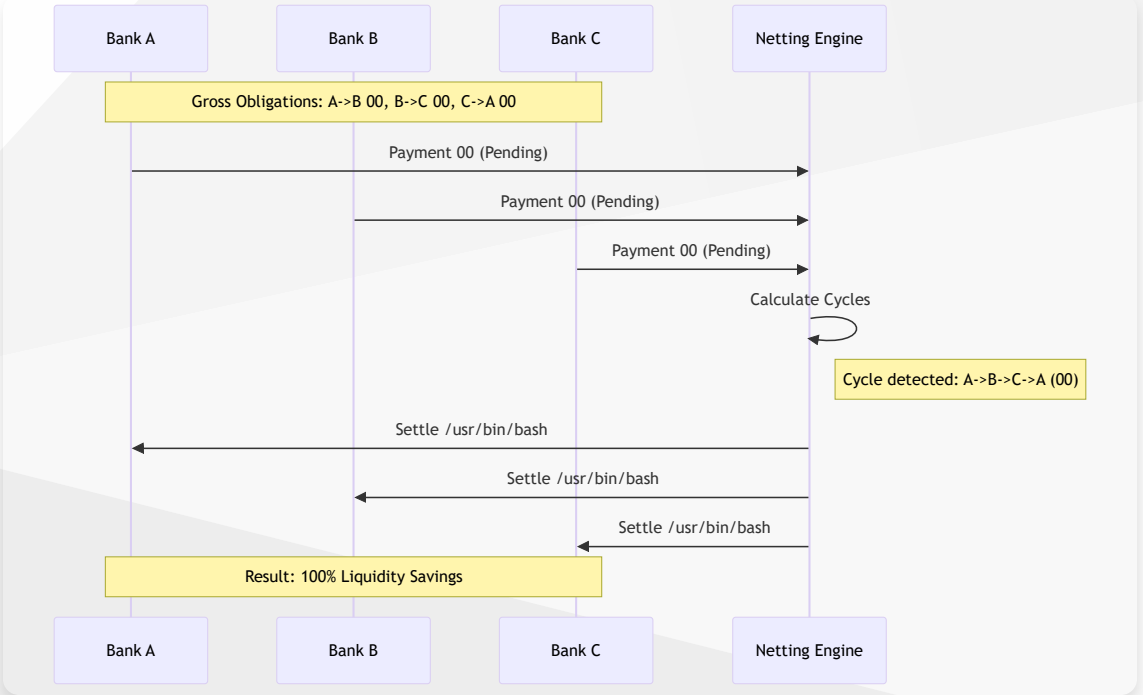
A portfolio-grade financial engine that implements **Real-Time Gross Settlement (RTGS)** and **Multilateral Netting** for Stablecoins.

## Hybrid Architecture

- ⚡ **Layer 1 (Off-Chain)**:
  10k+ TPS. Immediate clearing. Privacy-preserving compliance.
- ⛓️ **Layer 2 (On-Chain)**:
  Sepolia Testnet. Cryptographic proof of settlement.

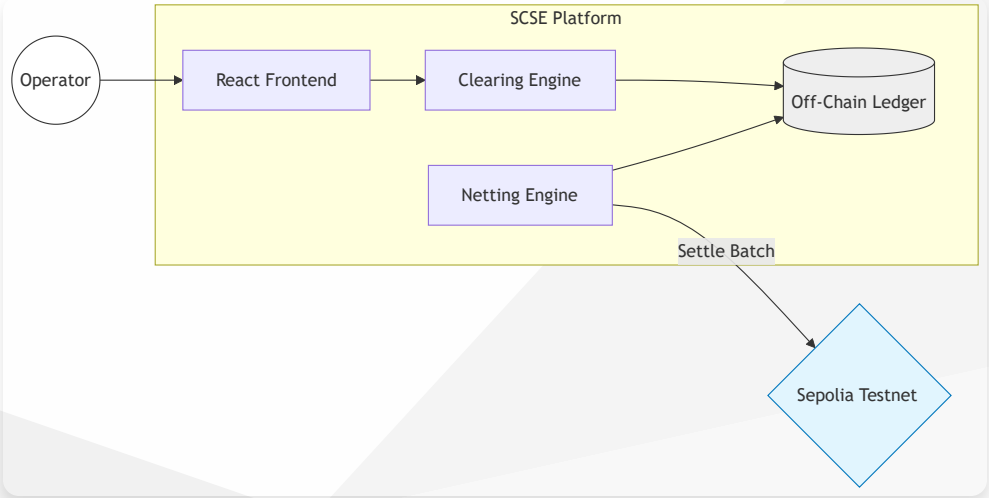
## 100% Liquidity Savings

Algorithms identify and resolve circular debts (A->B->C->A).



# Technical Architecture ⚙️

I built a full-stack simulation to demonstrate end-to-end payment lifecycles.




- **Frontend**: React, Mantine, Zustand.
- **Web3**: Wagmi, Viem, Solidity (Hardhat).
- **Logic**: Python (FastAPI/SQLAlchemy) adapted to client-side logic for the demo.

# The Cross-Border Vision 🌍

*So instead of Central Bank Money or Nostro/Vostro ledgers, can we use this?*

**YES. This replaces "Trapped Liquidity" with "Just-in-Time Liquidity".**

## 1. The Old Way (Correspondent Banking) 🏛️

- Bank A must keep **$10M** idle in Bank B's ledger (Nostro) just in case.
- **Cost**: That $10M earns 0% interest and cannot be used.

## 2. The SCSE Way (Netting + Stablecoins) 🚀

- Bank A and Bank B trade all day. Contract calculates Net = $0.
- **Result**: No money ever moves. No money is ever idle.
- **Settlement**: Only the *difference* moves on-chain, instantly, 24/7.

> ***Impact***: *Unlocks Trillions in global "Sleeping Money".*

# Engineering Philosophy & Impact 🌟

This project reflects my ability to bridge **Policy** and **Code**:

1. **Compliance-First**:
   Built with hooks for **Travel Rule (IVMS101)** compliance, ensuring regulatory readiness.
2. **Standards-Based**:
   Uses **ISO 20022** terminology (DebtorAgent, CreditorAgent) for interoperability.
3. **Real-World Reliability**:
   Implements proper **Double-Entry Accounting** logic, not just simple token transfers.

# About the Builder 👨‍💻

**Sivasubramanian Ramanathan**
*Product Owner | Fintech, RegTech & Digital Innovation*
*PMP | PSM II | PSPO II*

I specialize in taking messy, real-world complexity and structuring it into reliable products.

**Open for roles that sit between policy, technology, and stakeholder engagement.**

# Lets Connect 🤝

I am ready to bring this level of engineering rigor and product thinking to your team.

- 🌐 **Portfolio**: sivasub.com
- 💼 **LinkedIn**: linkedin.com/in/sivasub987
- 💻 **Code**: github.com/siva-sub/Stablecoin-Clearing-and-Settlement-Engine
- 📚 **Docs**: Full Documentation in /docs

**Live Demo**:
siva-sub.github.io/Stablecoin-Clearing-and-Settlement-Engine